7	**(a)**	Faraday's law of electromagnetic induction states that the e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage. Lenz's law states that the direction of the induced e.m.f. is such that it may produce an effect that opposes the change causing it.	
	(b)	**(i)**	Flux linkage = NBA [M1] $= 85 \times (\pi \times 10^{-3} \times 2.8) \times (\pi \times (1.6 \times 10^{-2})^2) = 6.0 \times 10^{-4}$ Wb [A1]
		(ii)	Flux change = $\Delta\Phi = 2 \times 6.00 \times 10^{-4}$ [M1] Induced e.m.f. = $\Delta\Phi/\Delta t$ = 0.00401 V ~ 4.0 mV [A1]
		(iii)	0 v for t = 0 to 0.3, t = 0.6 to 1.0 and t > 1.6 t [B1] 0.004 V for t = 0.3 to 0.6 s [B1] - 0.002 V for t = 1.0 to 1.6 s [B1]

Questions			**Answers**	**Marks**
8	**(a)**		Electrons have zero potential energy at infinity, and so less than	